

Cyclistic Bike-Share Analysis

How Do Annual Members and Casual Riders Use Cyclistic Bikes Differently?

A Case Study Report | Marketing Analytics Team

1. Business Task

Cyclistic's director of marketing, Lily Moreno, has asked the marketing analytics team to determine how annual members and casual riders use Cyclistic bikes differently. This analysis will inform a new marketing strategy aimed at converting casual riders into annual members, since Cyclistic's finance team has found that annual members generate significantly more revenue than casual riders. My task specifically is to identify usage patterns — such as ride length, ride frequency, time of day/week, and seasonality — that distinguish the two rider groups.

2. Data Sources

This analysis uses Cyclistic's historical trip data for the past 12 months, made available by Motivate International Inc. under Divvy's public data license. The data consists of 12 monthly CSV files, each containing individual ride records with fields including ride ID, bike type (classic or electric), ride start/end timestamps, start/end station names and IDs, start/end coordinates, and rider type (member or casual).

The dataset is reliable and credible under the ROCCC standard: it is original first-party operational data from the bike-share operator itself, comprehensive in covering all rides and stations, current (last 12 months), and publicly cited under an open license.

One limitation is that the data contains no personally identifiable information, so it is not possible to determine whether casual riders live within the service area or have purchased multiple single-ride passes — this rules out certain deeper customer-behavior questions. During initial inspection, a number of rows with missing start/end station names and coordinates were also noted, likely from bikes being redistributed by the operator rather than ridden between two stations; these were accounted for during cleaning.

3. Data Cleaning & Processing Documentation

The 12 monthly files were combined into a single table for analysis, totaling 5,932,349 rows. The following cleaning steps were then applied:

Cleaning Step	Rows Removed
Invalid timestamps (end time at or before start time)	29
Ride duration under 60 seconds (likely false starts/re-docks)	162,247
Duplicate ride IDs	35
Final cleaned dataset	5,770,032 rows

Two new fields were calculated to support analysis: `ride_length_seconds` (the duration of each ride) and `day_of_week` (the calendar day each ride started). A `ride_month` field was also derived to support seasonal analysis.

During analysis, a cluster of rides with unusually long durations (approximately 1,560 minutes, or roughly 26 hours) was identified, concentrated entirely on March 7, 2026, and affecting both casual and member riders. This pattern is consistent with a system-level event, such as a docking or outage issue, rather than genuine extended rides. These rows were retained in the dataset, as their effect on aggregate statistics is negligible, but are noted here as a known data anomaly.

Tools used: SQL (DB Browser for SQLite) for data combination, cleaning, and aggregation; Microsoft Excel for visualization.

4. Summary of Analysis

This analysis examined 5,770,032 cleaned rides from the past 12 months (2,036,289 from casual riders and 3,733,743 from members) to compare how annual members and casual riders use Cyclistic bikes differently.

Ride duration: Casual riders average 22.6 minutes per ride, nearly double the member average of 12.7 minutes. This suggests casual riders use bikes primarily for leisure, while members take shorter, more purposeful trips.

Day-of-week patterns: Casual ridership is heavily weekend-weighted, peaking on Saturday (428,281 rides) and Sunday (334,023 rides), with ride duration also longest on those two days (25.4 and 26.3 minutes respectively). Member ridership is highest on weekdays, particularly Tuesday through Thursday (595,000–606,000 rides each), and remarkably consistent in duration, holding between 12.2 and 12.7 minutes every day — a strong indicator of routine commuting behavior.

Seasonality: Casual ridership swings dramatically by season, from a low of 23,962 rides in January to a peak of 324,165 rides in August — roughly a 13x difference. Member ridership is far more stable across the year, ranging from 109,410 to 444,216 rides — about a 4x swing — meaning members continue riding through colder months at meaningfully higher rates than casual riders do.

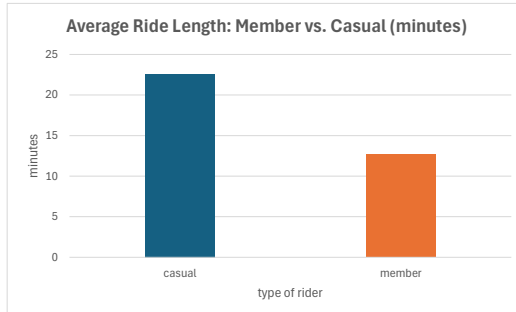
Bike type preference: Both groups show a similar preference for electric bikes over classic bikes (casual: 68.8% electric; member: 65.6% electric). This gap is small, making bike type a weaker differentiator than the findings above.

Overall conclusion: casual riders behave like weekend, seasonal, leisure users — riding longer, concentrated heavily on Saturdays and Sundays, and largely inactive outside the warmer months. Members behave like weekday commuters — shorter, highly consistent rides spread evenly across the work week, with meaningfully more ridership through the winter. This distinction directly shapes both the content and timing of a campaign aimed at converting casual riders into annual members.

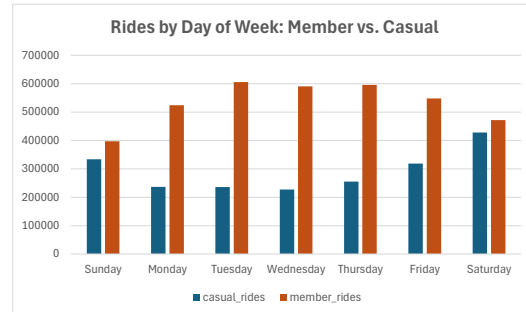
5. Supporting Visualizations & Key Findings

The dashboard below summarizes the four core comparisons underlying this analysis: average ride length, ride volume by day of week, seasonal ride patterns, and bike type preference — each split by rider type (member vs. casual).

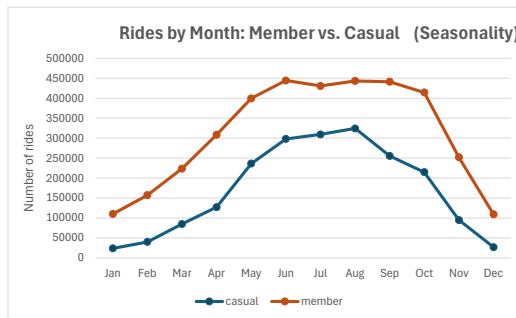
Cyclistic: How Annual Members and Casual Riders Use Bikes Differently



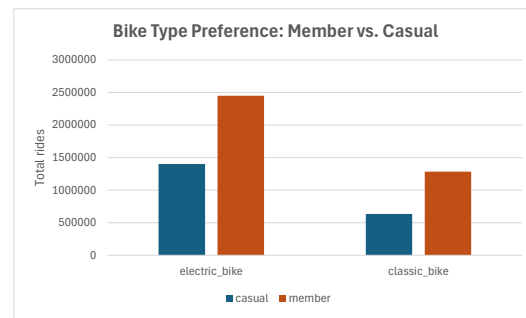
Casual riders take nearly 2x longer trips on average



Casual riders peak on weekends; members are steady weekday commuters.



Casual ridership swings 13x by season; members stay far more consistent.



Both groups prefer electric bikes at similar rates

Key takeaways from the dashboard:

- Casual riders take nearly 2x longer trips on average than members.
- Casual ridership peaks on weekends, while members ride steadily on weekdays — a clear commuter pattern.
- Casual ridership swings roughly 13x across the seasons (peaking in August); members remain far more consistent year-round.
- Both groups prefer electric bikes at similar rates (~65–69%), making this a weaker differentiator than the patterns above.

6. Top 3 Recommendations

1. Launch the campaign during peak casual season (May–August).

Casual ridership swings dramatically by season — from a low of 23,962 rides in January to a peak of 324,165 rides in August, roughly a 13x difference. Member ridership, by contrast, stays far more stable year-round. This tells us casual riders are most active, most visible, and most reachable in the warmer months. A membership-conversion campaign timed to launch in late spring and run through summer will reach casual riders while they're actively engaged with Cyclistic, rather than competing for attention during their off-season.

2. Shift campaign messaging from “commuting” to “weekend leisure.”

Casual riders show a clear weekend pattern: ride volume peaks on Saturday (428,281 rides) and Sunday (334,023 rides), and ride duration is also longest on those two days (25–26 minutes, versus 19–22 minutes mid-week). Members, meanwhile, show the opposite pattern — weekday-heavy, short, and remarkably consistent (12.2–12.7 minutes every day), reflecting routine commuting. Since casual riders don't currently behave like commuters, marketing messages centered on commute savings are unlikely to resonate. Messaging built around weekend value — for example, a “Weekend Explorer” angle, or promotions placed near stations with high weekend casual traffic — is more likely to match how these riders already use the service.

3. Introduce a lower-commitment membership tier to bridge casual riders into full membership.

Casual riders already average 22.6 minutes per ride, nearly double the member average of 12.7 minutes, showing they're getting real, sustained value from Cyclistic — they simply haven't committed to an annual plan. A discounted seasonal or weekend-only membership tier would let engaged casual riders experience membership benefits without the upfront commitment of a full year, making it easier to convert habitual users into long-term annual members over time.